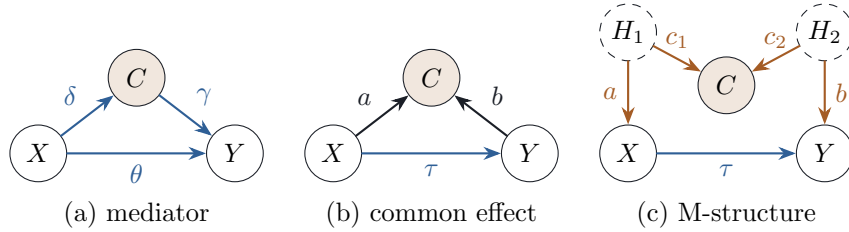


## 25.5 Bad controls: mediators and colliders

A regression that “controls for” many variables is not better for it. A control helps when it blocks a back-door path, costs at most precision when it lies on no path, and does harm when it lies on the causal path or when conditioning on it opens a closed path. Angrist and Pischke (2009) call the last two kinds **bad controls**. This section computes the damage in the linear models of Figure 25.5.1 and describes the related bias from selecting the sample on a collider.



**Figure 25.5.1.** Three bad controls  $C$  (shaded). (a)  $C$  is a mediator. (b)  $C$  is a common effect of  $X$  and  $Y$ . (c)  $C$  is measured before  $X$  but is a collider on the back-door path through the unrecorded  $H_1$  and  $H_2$ .

### 25.5.1 Three computations

#### Theorem 25.5.1 — Bad controls in linear models

In each model below the disturbances  $U_X, U_Y, U_C$  (and, in (c), the unrecorded variables  $H_1, H_2$ ) are independent with mean zero and variances  $\sigma_X^2, \sigma_Y^2, \sigma_C^2 > 0$  (and 1). Let  $\tau$  be the total effect of  $X$  on  $Y$ . In every case the slope of  $L(Y | X)$  equals  $\tau$ , so the simple regression is consistent, while the coefficient  $\beta_{X|C}$  of  $X$  in  $L(Y | X, C)$  is as follows.

- (a) **Mediator:**  $X = U_X$ ,  $C = \delta X + U_C$ ,  $Y = \theta X + \gamma C + U_Y$ . Here  $\tau = \theta + \gamma\delta$  and  $\beta_{X|C} = \theta$ : adjustment removes the indirect effect  $\gamma\delta$ .
- (b) **Common effect:**  $X = U_X$ ,  $Y = \tau X + U_Y$ ,  $C = aX + bY + U_C$ . Then

$$\beta_{X|C} = \tau - \frac{(a + b\tau) b \sigma_Y^2}{b^2 \sigma_Y^2 + \sigma_C^2}. \quad (25.5.1)$$

- (c) **M-structure:**  $X = aH_1 + U_X$ ,  $C = c_1H_1 + c_2H_2 + U_C$ ,  $Y = \tau X + bH_2 + U_Y$ . Then

$$\beta_{X|C} = \tau - \frac{a b c_1 c_2}{D}, \quad (25.5.2)$$

$$D = (a^2 + \sigma_X^2)(c_1^2 + c_2^2 + \sigma_C^2) - a^2 c_1^2 > 0.$$

*Proof.*

- (a)  $Y - \theta X - \gamma C = U_Y$  is uncorrelated with  $X$  and  $C$ , so  $L(Y | X, C) = \theta X + \gamma C$ . Without  $C$ ,  $Y = (\theta + \gamma\delta)X + (\gamma U_C + U_Y)$ , and the bracket is uncorrelated with  $X$ , so the slope is  $\tau$ .
- (b) Without  $C$ ,  $U_Y$  is uncorrelated with  $X$  and the slope is  $\tau$ . With  $C$ , write  $E =$

$C - (a + b\tau)X = bU_Y + U_C$ , which is uncorrelated with  $X$ . The span of  $1, X, C$  is the span of  $1, X, E$ , with  $E$  orthogonal to  $1$  and  $X$ , so  $L(U_Y | X, C) = L(U_Y | X) + L(U_Y | E) = 0 + \lambda E$ , where  $\lambda = \text{Cov}(U_Y, E) / \text{Var}(E) = b\sigma_Y^2 / (b^2\sigma_Y^2 + \sigma_C^2)$ . Hence  $L(Y | X, C) = \tau X + \lambda C - \lambda(a + b\tau)X$ , whose coefficient of  $X$  is [Eq. 25.5.1](#).

- (c) Without  $C$ ,  $bH_2 + U_Y$  is uncorrelated with  $X$  and the slope is  $\tau$ . With  $C$ ,  $L(Y | X, C) = \tau X + bL(H_2 | X, C)$ , since  $U_Y$  is uncorrelated with  $X$  and  $C$ . The coefficients of  $L(H_2 | X, C)$  solve the population equations of Theorem 6.11.1, with covariance matrix of  $(X, C)$  equal to

$$\begin{pmatrix} a^2 + \sigma_X^2 & ac_1 \\ ac_1 & c_1^2 + c_2^2 + \sigma_C^2 \end{pmatrix}$$

and right-hand side  $(\text{Cov}(X, H_2), \text{Cov}(C, H_2)) = (0, c_2)$ . The determinant is  $D$ , positive since the matrix is a positive definite covariance matrix, and the first coefficient is  $-ac_1c_2/D$ . Multiply by  $b$ .

■

In (a) the adjusted coefficient is the **direct effect**, with the mediator held fixed by intervention. That may be of interest, but only if nothing confounds the mediator and the outcome ([Exercise \(B2\)](#)); a policy decision needs  $\tau$  ([Section 18.1](#) gave the experimental version).

In (b) the bias exists even when  $\tau = 0$ : adjusting for a consequence of both variables produces an “effect”  $-ab\sigma_Y^2 / (b^2\sigma_Y^2 + \sigma_C^2)$  where there is none, negative when  $a$  and  $b$  have the same sign. This is the collider of Proposition 25.3.3(c).

In (c) the control is measured before  $X$ , so the advice to adjust only for pre-treatment variables does not exclude it, yet adjustment opens the back-door path  $X \leftarrow H_1 \rightarrow C \leftarrow H_2 \rightarrow Y$ . This **M-bias**, named after the graph’s shape, was pointed out by Greenland, Pearl and Robins (1999). It is a product of four path coefficients over  $D$  and tends to be small unless all four are large (Ding and Miratrix 2015). When the collider is also a confounder, with arrows  $C \rightarrow X$  and  $C \rightarrow Y$ , neither choice removes the bias; Ding and Miratrix call this butterfly bias.

### Example. (Three bad controls by simulation)

Take, with unit variances throughout:

- a mediator with  $\theta = 0.5$ ,  $\delta = 1.2$ ,  $\gamma = 1$ , so that  $\tau = 1.7$  and  $\beta_{X|C} = 0.5$ ;
- a common effect with  $\tau = 1$ ,  $a = 0.5$ ,  $b = 1$ , for which [Eq. 25.5.1](#) gives  $\beta_{X|C} = 1 - (1.5)(1)/2 = 0.25$ ;
- an M-structure with  $\tau = 1$  and  $a = b = c_1 = c_2 = 1.5$ , for which  $D = \frac{13}{4} \cdot \frac{11}{2} - \frac{81}{16} = \frac{205}{16} = 12.8125$  and  $\beta_{X|C} = 1 - \frac{81}{205} = 0.605$ .

Over 4000 samples of  $n = 400$ , the least squares coefficients of  $X$  without and with  $C$  average 1.700 and 0.499 for the mediator, 1.000 and 0.251 for the common effect, and 0.999 and 0.605 for the M-structure ([Figure 25.5.2](#)). In the M-structure the adjusted estimate is also slightly *less* variable (standard deviation 0.046 against 0.050), a reminder that a smaller standard error says nothing about bias.

```
import numpy as np

rng = np.random.default_rng(2505)
```

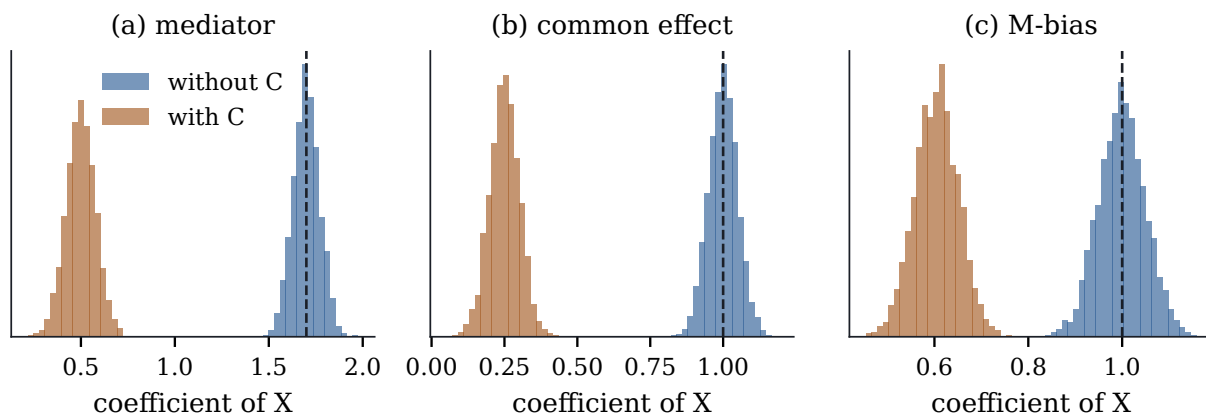


Figure 25.5.1: Sampling distributions of the least squares coefficient of  $X$  without (blue) and with (brown) adjustment for  $C$ , in the three models of [Example \(Three bad controls by simulation\)](#); 4000 samples of size 400. The dashed line is the total effect  $\tau$ .

```
def mediator(n):
    # X -> C -> Y and X -> Y; total effect 0.5 + 1.2 * 1.0 = 1.7
    x = rng.normal(size=n); c = 1.2 * x + rng.normal(size=n)
    return x, c, 0.5 * x + 1.0 * c + rng.normal(size=n)

def collider(n):
    # X -> Y, and X -> C <- Y; total effect 1
    x = rng.normal(size=n); y = 1.0 * x + rng.normal(size=n)
    return x, 0.5 * x + 1.0 * y + rng.normal(size=n), y

def m_bias(n):
    # X <- H1 -> C <- H2 -> Y, and X -> Y; total effect 1
    h1, h2 = rng.normal(size=(2, n))
    x = 1.5 * h1 + rng.normal(size=n)
    c = 1.5 * h1 + 1.5 * h2 + rng.normal(size=n)
    return x, c, 1.0 * x + 1.5 * h2 + rng.normal(size=n)

def slopes(x, c, y):
    """Coefficient of x without and with adjustment for c."""
    one = np.ones(len(x))
    b_short = np.linalg.lstsq(np.column_stack([one, x]), y, rcond=None)[0][1]
    b_long = np.linalg.lstsq(np.column_stack([one, x, c]), y, rcond=None)[0][1]
    return b_short, b_long
```

[Run this code online](#)

```
n, reps = 400, 500
for name, model in [("mediator", mediator), ("collider", collider), ("M-bias",
    m_bias)]:
    est = np.array([slopes(*model(n)) for _ in range(reps)])
    print(f"{name:9s} mean without C {est[:, 0].mean():.3f}    mean with C {est[:,
    1].mean():.3f}")
```

[Run this code online](#)

## 25.5.2 Post-treatment variables in general

A variable affected by  $X$  removes part of the effect if it lies on a directed path to  $Y$ , and opens a path if  $Y$  or a cause of  $Y$  also affects it; its descendants act partially (Exercise (B1)). A descendant of  $X$  with no other connection to  $Y$  costs only precision (Exercise (B1)). “Adjust only for pre-treatment variables” is sound in randomized experiments (Section 25.6); in observational data it is neither sufficient, by the M-structure, nor necessary, since a later measurement can proxy an unrecorded confounder. The graph, not the timing, decides. Adjusting for a pure cause of  $X$  can even *increase* the bias from an unrecorded confounder (Exercise (B3)); Pearl (2010) calls such variables bias amplifiers.

## 25.5.3 Selection on a collider

A sample selected on a common effect of  $X$  and  $Y$  is conditioned on a collider without any adjustment. Berkson (1946) saw two diseases, independent in the population, associated among hospital patients, since either brings a patient to hospital; Elwert and Winship (2014) survey this *endogenous selection bias*. In the common-effect model of Example (Three bad controls by simulation), observing only the units with  $C > 1$  (about 0.31 of them) moves the slope of  $Y$  on  $X$  from 1.002 to 0.584.

```
x, c, y = collider(200_000)
keep = c > 1                                # only units with a large C
are observed
slope_all = np.polyfit(x, y, 1)[0]
slope_sel = np.polyfit(x[keep], y[keep], 1)[0]
print(f"slope of Y on X: all units {slope_all:.3f}, units with C > 1 {slope_sel:.3f}")
```

[Run this code online](#)

*Warning.* A variable belongs in a causal regression because of its position in the graph, not because it improves  $R^2$  or is “significant”; predictive model selection (Chapter 29) readily selects mediators and colliders.

## 25.5.4 Exercises

### 25.5.4.1 A. Check your understanding

#### Exercise A1.

A study of the effect of a training programme ( $X$ ) on earnings two years later ( $Y$ ) considers adjusting for: (i) education before the programme; (ii) the job obtained six months after the programme; (iii) whether the person was still living in the region at the end of the study, which depends on both training and earnings; (iv) the day of the week on which the person applied. Classify each as a confounder, mediator, collider or irrelevant, under a plausible graph, and say which should be adjusted for.

### 25.5.4.2 B. Practice

**Exercise B1.**

Let  $X = U_X$ ,  $Y = \tau X + U_Y$  and  $D = dX + U_D$ , where  $D$  is affected by  $X$  but has no connection with  $Y$ . Show that the coefficient of  $X$  in  $L(Y | X, D)$  is  $\tau$ , and that the asymptotic variance of the least squares coefficient is larger with  $D$  than without it by the factor  $(d^2\sigma_X^2 + \sigma_D^2)/\sigma_D^2$ .

**Exercise B2.**

In the mediator model add an unrecorded  $U$ , with variance  $\sigma_U^2$ , that affects both the mediator and the outcome:  $C = \delta X + gU + U_C$ ,  $Y = \theta X + \gamma C + hU + U_Y$ . Show that

$$\beta_{X|C} = \theta - \frac{\delta g h \sigma_U^2}{g^2 \sigma_U^2 + \sigma_C^2},$$

so that adjusting for the mediator does not even recover the direct effect  $\theta$ .

*Solution.*  $Y = \theta X + \gamma C + hU + U_Y$  and  $U_Y$  is uncorrelated with  $(X, C)$ , so  $\beta_{X|C} = \theta + h\kappa$ , where  $\kappa$  is the coefficient of  $X$  in  $L(U | X, C)$ . Put  $E = C - \delta X = gU + U_C$ , uncorrelated with  $X$ . As in the proof of [Theorem 25.5.1\(b\)](#),  $L(U | X, C) = L(U | E) = \lambda(C - \delta X)$  with  $\lambda = g\sigma_U^2/(g^2\sigma_U^2 + \sigma_C^2)$ , so  $\kappa = -\lambda\delta$ . The mediator is a collider on the path  $X \rightarrow C \leftarrow U \rightarrow Y$ , which adjustment opens.

**Exercise B3.**

Let  $U$  be unrecorded,  $X = \pi I + gU + U_X$  and  $Y = \tau X + hU + U_Y$ , with  $I, U, U_X, U_Y$  independent,  $\text{Var } I = \text{Var } U = 1$  and  $\text{Var } U_X = \sigma^2$ . Show that the slope of  $L(Y | X)$  is  $\tau + gh/(\pi^2 + g^2 + \sigma^2)$ , and that the coefficient of  $X$  in  $L(Y | X, I)$  is  $\tau + gh/(g^2 + \sigma^2)$ . Adjusting for the instrument  $I$  enlarges the bias.

*Solution.*  $\text{Cov}(X, Y) = \tau \text{Var } X + gh$  and  $\text{Var } X = \pi^2 + g^2 + \sigma^2$ , which gives the first slope. For the second, by Proposition 14.5.2 the coefficient is  $\sigma_{YX \cdot I}/\sigma_{XX \cdot I}$ . Removing the linear effect of  $I$  leaves  $X - \pi I = gU + U_X$ , with variance  $g^2 + \sigma^2$ , and  $Y - L(Y | I) = \tau(gU + U_X) + hU + U_Y$ , so  $\sigma_{YX \cdot I} = \tau(g^2 + \sigma^2) + gh$ . Divide. The confounded covariance  $gh$  is the same in both, but adjustment divides it by a smaller variance.

**25.5.4.3 C. Going deeper****Exercise C1.**

(*Butterfly bias.*) Let  $H_1, H_2, U_C, U_X, U_Y$  be independent with mean zero and variance 1, and let

$$C = H_1 + H_2 + U_C, \quad X = H_1 + C + U_X, \quad Y = \tau X + H_2 + C + U_Y,$$

so that  $C$  is at once a collider on  $H_1 \rightarrow C \leftarrow H_2$  and a common cause of  $X$  and  $Y$ .

- (a) List the back-door paths from  $X$  to  $Y$  and say which are open when  $C$  is not conditioned on, and which when it is. Conclude that neither  $\emptyset$  nor  $\{C\}$  satisfies

the back-door criterion (Definition 25.4.1).

- (b) Show that the slope of  $L(Y \mid X)$  is  $\tau + \frac{5}{7}$  and that the coefficient of  $X$  in  $L(Y \mid X, C)$  is  $\tau - \frac{1}{5}$ . Adjusting for  $C$  reduces the bias and reverses its sign, so the two estimates bracket  $\tau$ .
- (c) Now change one arrow, replacing  $C = H_1 + H_2 + U_C$  by  $C = H_1 - H_2 + U_C$  and keeping everything else. Show that the two biases become  $\frac{3}{7}$  and  $\frac{1}{5}$ : both positive, so the two estimates no longer bracket the truth.

There is therefore no “adjust and compare” rule that turns two biased estimates into bounds.

*Solution.*

- (a) Write the paths that leave  $X$  by an arrow into  $X$ . They are  $X \leftarrow C \rightarrow Y$ ,  $X \leftarrow H_1 \rightarrow C \rightarrow Y$ ,  $X \leftarrow C \leftarrow H_2 \rightarrow Y$  and  $X \leftarrow H_1 \rightarrow C \leftarrow H_2 \rightarrow Y$ . On the first three,  $C$  is a non-collider, so they are open when  $C$  is free and blocked when it is conditioned on. On the fourth,  $C$  is a collider, so that path is blocked when  $C$  is free and opened when it is conditioned on. Condition (i) is not at issue, since  $C$  is no descendant of  $X$ ; it is (ii) that fails for both candidate sets, because every choice leaves one of the four paths open.
- (b) Substituting,  $X = 2H_1 + H_2 + U_C + U_X$  and  $Y = \tau X + R$  with  $R = H_1 + 2H_2 + U_C + U_Y$ . All five disturbances are uncorrelated with variance one, so  $\text{Var}(X) = 4 + 1 + 1 + 1 = 7$ ,  $\text{Cov}(X, R) = 2 + 2 + 1 = 5$ ,  $\text{Var}(C) = 3$ ,  $\text{Cov}(X, C) = 2 + 1 + 1 = 4$  and  $\text{Cov}(R, C) = 1 + 2 + 1 = 4$ . The slope of  $L(Y \mid X)$  is  $\tau + \text{Cov}(X, R) / \text{Var}(X) = \tau + \frac{5}{7}$  (Theorem 6.11.1). For the adjusted coefficient,  $U_Y$  is uncorrelated with  $(X, C)$ , so the coefficient of  $X$  in  $L(Y \mid X, C)$  is  $\tau + k_1$ , where  $(k_1, k_2)$  solves the population normal equations

$$\begin{pmatrix} 7 & 4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}.$$

The determinant is 5 and  $k_1 = (5 \cdot 3 - 4 \cdot 4) / 5 = -\frac{1}{5}$ .

- (c) Now  $X = 2H_1 - H_2 + U_C + U_X$  with  $\text{Var}(X) = 7$  again, and  $R = H_2 + C + U_Y = H_1 + U_C + U_Y$ , so  $\text{Cov}(X, R) = 2 + 1 = 3$  and the unadjusted bias is  $\frac{3}{7}$ . Also  $\text{Var}(C) = 3$ ,  $\text{Cov}(X, C) = 2 + 1 + 1 = 4$  and  $\text{Cov}(R, C) = 1 + 1 = 2$ , so the same matrix now faces the right-hand side  $(3, 2)^\top$  and  $k_1 = (3 \cdot 3 - 4 \cdot 2) / 5 = \frac{1}{5}$ .

The sign of the M-bias that conditioning opens is the sign of a product of four path coefficients (Eq. 25.5.2), while the sign of the confounding that conditioning closes is set by the arrows out of  $C$ ; changing one arrow changes the first and not the second. Bracketing in (b) is an accident of the numbers, and nothing in the graph guarantees it.

### Exercise C2.

(A bad control that does no harm.) Let  $H$  be unrecorded,  $U_X, U_C, U_Y$  independent of it and of each other,  $\text{Var}(H) = \text{Var}(U_X) = 1$ ,  $\text{Var}(U_C) = \sigma_C^2 > 0$ , and

$$X = H + U_X, \quad Y = \tau X + H + U_Y, \quad C = pH + qX + U_C.$$

The control  $C$  is measured after the treatment: it is a descendant of  $X$  and a collider

on  $X \rightarrow C \leftarrow H \rightarrow Y$ , so the advice of this section forbids it twice over.

- (a) Show that the slope of  $L(Y | X)$  is  $\tau + \frac{1}{2}$ .
- (b) Show that the coefficient of  $X$  in  $L(Y | X, C)$  is

$$\tau + \frac{\sigma_C^2 - pq}{p^2 + 2\sigma_C^2}.$$

- (c) Deduce that the adjusted coefficient is exactly  $\tau$  when  $\sigma_C^2 = pq$ , for instance at  $p = q = \sigma_C^2 = 1$ , and check the two extreme cases  $q = 0$  and  $p = 0$  against [Exercise \(B1\)](#) and against the idea of a proxy for  $H$ .

Why does this not overturn the advice?

*Solution.*

- (a)  $Y = \tau X + R$  with  $R = H + U_Y$ , and  $\text{Var}(X) = 2$ ,  $\text{Cov}(X, R) = 1$ , so the slope is  $\tau + \frac{1}{2}$ .
- (b) Since  $U_Y$  is uncorrelated with  $(X, C)$ , the coefficient of  $X$  in  $L(Y | X, C)$  is  $\tau + k_1$ , with  $(k_1, k_2)$  the coefficients of  $L(H | X, C)$ . Write  $C = (p + q)H + qU_X + U_C$ . Then  $\text{Var}(X) = 2$ ,  $\text{Cov}(X, C) = p + 2q$ ,  $\text{Var}(C) = (p + q)^2 + q^2 + \sigma_C^2$ ,  $\text{Cov}(H, X) = 1$  and  $\text{Cov}(H, C) = p + q$ . The determinant of the covariance matrix of  $(X, C)$  is

$$D = 2\{(p + q)^2 + q^2 + \sigma_C^2\} - (p + 2q)^2 = p^2 + 2\sigma_C^2 > 0,$$

and solving the two equations for  $k_1$  gives  $k_1 = \{(p + q)^2 + q^2 + \sigma_C^2 - (p + 2q)(p + q)\}/D = (\sigma_C^2 - pq)/D$ , since the brace equals  $(p + q)\{(p + q) - (p + 2q)\} + q^2 + \sigma_C^2 = \sigma_C^2 - pq$ .

- (c) At  $p = q = \sigma_C^2 = 1$  the numerator vanishes and the adjusted coefficient is exactly  $\tau$ , while the unadjusted one is  $\tau + \frac{1}{2}$ . At  $q = 0$ ,  $C = pH + U_C$  is a pre-treatment proxy for the confounder and the bias is  $\sigma_C^2/(p^2 + 2\sigma_C^2) < \frac{1}{2}$ , tending to 0 as  $\sigma_C^2 \rightarrow 0$ : a perfect proxy is as good as  $H$  itself. At  $p = 0$ ,  $C = qX + U_C$  is a pure descendant of  $X$  and the bias is  $\sigma_C^2/(2\sigma_C^2) = \frac{1}{2}$ , unchanged, which is [Exercise \(B1\)](#) again.

The advice survives because the cancellation is not a property of the graph. Conditioning on  $C$  opens the path  $X \rightarrow C \leftarrow H \rightarrow Y$ , whose contribution is  $-pq/D$ , and shrinks the confounding along  $X \leftarrow H \rightarrow Y$  to  $\sigma_C^2/D$ ; the two are equal only on the surface  $\sigma_C^2 = pq$  in the parameter space, and  $p$ ,  $q$  and  $\sigma_C^2$  all involve the unrecorded  $H$ , so no data tell the analyst where on that surface the truth lies. A rule that forbids the control is right in the sense that matters: it is right for every graph, while the exception holds for a null set of the parameters and cannot be recognized.